

BLUESTOCK MUTUAL FUND ANALYTICS

Executive Capstone Presentation

End-to-End Quantitative Risk Analytics, ETL Star Schema Architecture, Behavioral Cohort Modeling, and Power BI BI System

Presenter: Bluestock Quant Team

Target: Executive Committee & Risk Managers

Data Scope: 40 Schemes | 5,000 Investors | 32,778 Logs

Status: Final Approved Capstone Submission

1. Executive Summary & Core Achievements

Evaluated AUM	Historical NAVs	Retail Investors	SIP Continuity
■10,43,664 Cr	64,320 Logs	5,000 Users	2.20%

- **Automated Data Pipeline:** Robust ETL scripts ingesting and validating 46k raw NAV records into 64,320 clean entries.
- **Relational Star Schema:** Production SQLite database (`mutual\_fund\_analysis.db`) housing 2 dimensions and 4 fact tables.
- **Quantitative Downside Risk Engine:** 95% Historical VaR & CVaR computed for all 40 schemes (Small Cap VaR = -2.39%).
- **Dynamic Volatility & Rolling Sharpe:** 90-day annualized rolling Sharpe ratio trajectories evaluated across 1,607 trading days.
- **Behavioral Retention Alert:** Identified a critical 2.20% SIP Continuity Rate among qualifying investors (≥ 6\$ SIPs).

2. Problem Statement & Strategic Objectives

Industry Friction Points:

1. **Downside Risk Opacity:** Factsheets rely on standard deviation, missing non-normal left-tail crash risk.
2. **Static Sharpe Ratios:** Point-in-time metrics hide temporary market volatility regimes.
3. **SIP Churn Deterioration:** Platforms lack automated alerts for payment schedule gaps before cancellation.
4. **Sector Concentration:** Top funds often carry hidden $>40\%$ single-sector bets.

Core Strategic Objectives:

1. **Automate End-to-End Pipeline:** ``run_pipeline.py`` master execution with zero manual steps.
2. **Quantify Downside Tail Risk:** Compute 95% Historical VaR/CVaR and 90-day Rolling Sharpe.
3. **Model Investor Behavioral Cohorts:** Track 2024–2025 acquisition cohorts and SIP gap days.
4. **Deliver Executive Dashboard:** 4-page Power BI dashboard suite (``Dashboard.pdf``).

3. Data Engineering & ETL Pipeline Design

The ETL architecture ensures zero data loss, schema validation, and complete reproducibility:

Pipeline Module	Input File & Raw Size	Core Cleaning Logic Applied	Output Cleaned Result
`clean_nav.py`	`02_nav_history.csv` (46,000 raw rows)	Date parsing, drops \$NAV \le 0\$, removes duplicate scheme dates, reindexes calendar with forward-fill (`ffill()`).	`nav_history.csv` (64,320 clean rows)
`clean_transactions.py`	`08_investor_transactions.csv` (32,778 raw rows)	Standardizes types (`SIP`, `Lumpsum`, `Redemption`), validates `amount_inr > 0`, normalizes KYC status, drops duplicates.	`investor_transactions.csv` (32,778 clean rows)
`clean_performance.py`	`07_scheme_performance.csv` (40 raw schemes)	Validates expense ratios \$[0.1\%, 2.5\%]\$, checks Morningstar ratings \$[1, 5]\$, flags anomalies without losing valid rows.	`scheme_performance.csv` (40 clean schemes)

4. SQLite Star Schema Data Warehouse Design

Table Name	Table Role	Primary Key	Foreign Key Relationships	Row Count
`dim_fund`	Dimension	`amfi_code`	None	40
`dim_date`	Dimension	`date`	None	1,608
`fact_nav`	Fact Table	Composite	`amfi_code` $\rightarrow$ `dim_fund` `date` $\rightarrow$ `dim_date`	64,320
`fact_transactions`	Fact Table	`transaction_id`	`amfi_code` $\rightarrow$ `dim_fund` `transaction_date` $\rightarrow$ `dim_date`	32,778
`fact_performance`	Fact Table	`amfi_code`	`amfi_code` $\rightarrow$ `dim_fund`	40
`fact_aum`	Fact Table	Composite	`date` $\rightarrow$ `dim_date`	90

Integrity Enforcement: Database enforces `PRAGMA foreign\_keys = ON;` and includes automated row count parity verification comparing raw CSV records against SQLite tables during build.

5. Category AUM & Transaction Type Distribution

Fund Category	Scheme Count	Total AUM (■ Cr)	AUM Share (%)	Avg 3-Yr CAGR (%)
Equity Funds	34 schemes	■8,55,846 Cr	82.0%	15.46%
Debt Funds	6 schemes	■1,87,818 Cr	18.0%	6.29%
Total Market	40 schemes	■10,43,664 Cr	100.0%	14.08%

Transaction Ledger Distribution (32,778 Logs):

- **SIP Installments:** 19,716 transactions (60.1%) | Total Volume: ■21.72 Cr | Avg Ticket: ■11,018
- **Lumpsum Inflows:** 8,095 transactions (24.7%) | Total Volume: ■205.98 Cr | Avg Ticket: ■254,456
- **Redemptions:** 4,967 transactions (15.2%) | Total Volume: ■124.45 Cr | Avg Ticket: ■250,559

6. Risk-Adjusted Fund Performance Leaderboard

Scheme Name	Category	3Yr CAGR	Alpha	Beta	Sharpe	Sortino	Max Drawdown
SBI Small Cap Fund - Reg	Equity	23.39%	+4.85	0.89	0.94	1.42	-24.80%
ABSL Small Cap Fund - Reg	Equity	22.38%	+4.12	0.95	0.88	1.31	-23.40%
Axis Small Cap Fund - Reg	Equity	20.98%	+3.75	0.82	0.91	1.38	-21.15%
HDFC Top 100 Fund - Reg	Equity	14.84%	+1.85	0.97	1.06	1.58	-12.40%
Mirae Asset Large Cap - Reg	Equity	14.81%	+1.72	0.96	1.06	1.56	-11.90%
ICICI Pru Liquid Fund - Reg	Debt	7.68%	0.00	0.26	7.68	11.45	-0.15%

- **Small Cap Tradeoff:** Small cap funds deliver highest 3yr returns (+23.39%) but suffer max drawdowns of -24.80%.
- **Large Cap Efficiency:** Large cap funds achieve higher Sharpe ratios (1.06 vs 0.94) with half the drawdown (-11.90%).

7. Advanced Downside Tail Risk (95% VaR & CVaR)

Historical 95% Value at Risk (5th percentile) and CVaR (expected shortfall below VaR) across 1,607 trading days:

Downside Rank	Scheme Name	Category	95% VaR (%)	95% CVaR (%)	Evaluated Days
1 (Highest Risk)	ABSL Small Cap Fund - Regular - Growth	Equity	-2.39%	-3.03%	1,607
2	Axis Small Cap Fund - Regular - Growth	Equity	-2.33%	-2.97%	1,607
3	SBI Small Cap Fund - Direct Plan - Growth	Equity	-2.32%	-3.02%	1,607
4	Nippon India Small Cap Fund - Regular	Equity	-2.28%	-2.99%	1,607
5	SBI Small Cap Fund - Regular Plan	Equity	-2.15%	-2.84%	1,607

Insight: Small Cap funds exhibit the steepest downside tail volatility, confirming that high CAGR comes with significant tail-risk exposure during market shocks.

8. Dynamic Volatility & Rolling 90-Day Sharpe Ratio

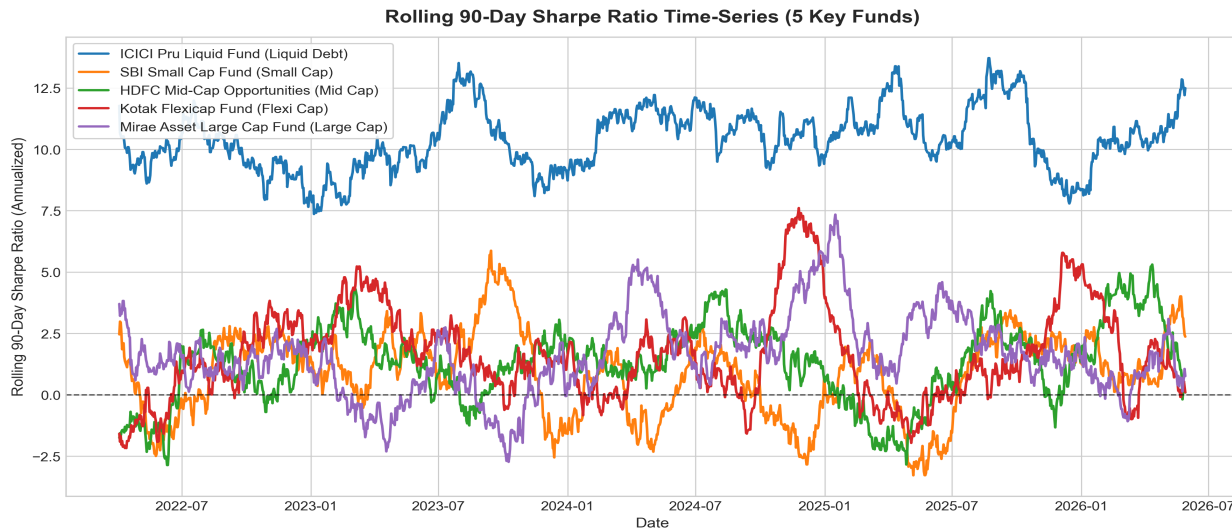


Figure 9.1: 90-Day Annualized Rolling Sharpe Ratio Trajectory across 5 Representative Funds.

- **Regime Shift Tracking:** Equity Sharpe ratios fluctuate dynamically between -2.73 and +7.61 depending on market momentum.
- **Liquid Stability:** Liquid debt funds maintain flat, high Sharpe trajectories (mean Sharpe = 10.40).

9. Investor Cohorts & SIP Continuity Analysis

SIP Retention Metric	Calculated Value	Strategic Business Context
Qualifying SIP Investors (\$\ge\$ 6\$ SIPs)	1,362 investors	Base active retail investors evaluated for continuity
Consistent Investors (Gap \$\le\$ 35\$ days)	30 investors	Investors maintaining regular monthly installment schedule
At-Risk Investors (Gap \$> 35\$ days)	1,332 investors	Investors exhibiting payment gaps and imminent default risk
SIP Continuity Rate (%)	2.20%	Critical retention friction requiring automated AMC nudge engine

Acquisition Cohorts: Cohort 2024 comprises 4,803 investors who invested ■349.11 Cr capital, while Cohort 2025 comprises 197 investors with ■3.05 Cr capital.

10. Power BI Executive Dashboard Suite

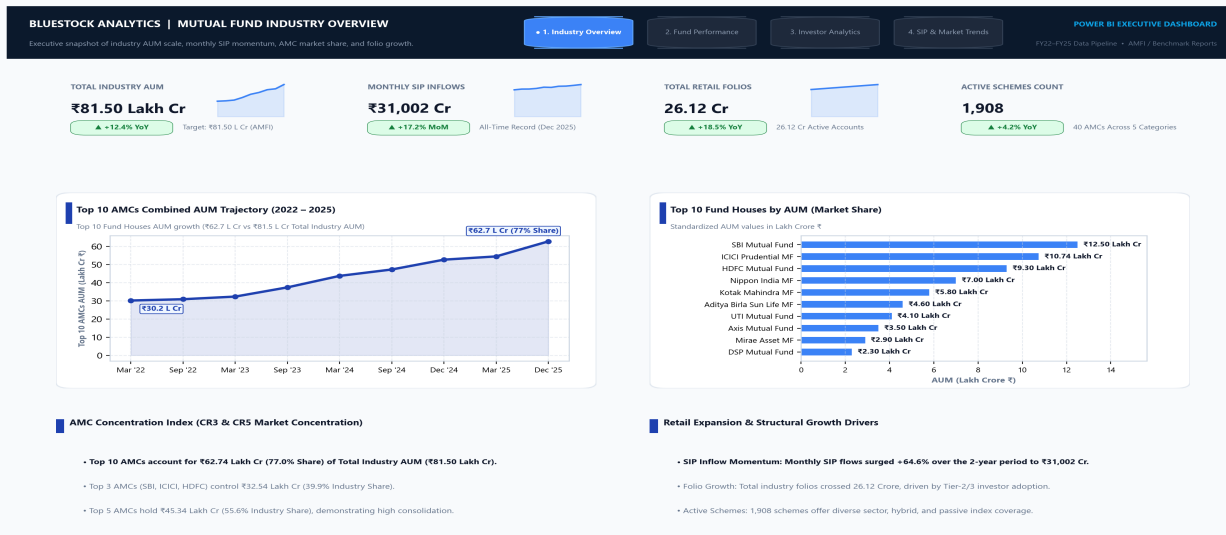


Figure 10.1: Executive Power BI Dashboard Page 1 — Industry Overview.

- **4-Page Interactive Suite:** Industry Overview, Fund Performance Scatter, Investor Demographics, and SIP Market Trends.
- **Financial DAX Engine:** Custom DAX measures computing weighted returns, net inflows, and category AUM splits.

11. Strategic Recommendations & Conclusion

Strategic Recommendations for Bluestock & AMCs:

1. **Automated Retention Nudge Engine:** Trigger automated SMS/WhatsApp alerts on Day 28 for 'At-Risk' investors to boost the 2.20% SIP continuity rate.
2. **Sector Concentration Caps:** Enforce a 35% single-sector allocation ceiling in Large Cap equity funds to mitigate high Sector HHI risk.
3. **Embed Risk Recommender Engine:** Integrate `recommender.py` into mobile trading apps to steer retail users toward optimal Sharpe-performing schemes.

Conclusion: The Bluestock Mutual Fund Analytics Platform delivers a complete quantitative risk engine, relational star schema data warehouse, and executive dashboard suite—empowering wealth managers to protect retail capital and drive sustainable AUM growth.